DAY 04 · THEORY HANDOUT

Simple charts with Python

Matplotlib and Seaborn, using small everyday examples.

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Matplotlib and Seaborn: why both?

A library is ready code we can use in our own program.

Matplotlib

Draws common charts.

Adds titles and axis labels.

Seaborn

Uses Matplotlib to draw.

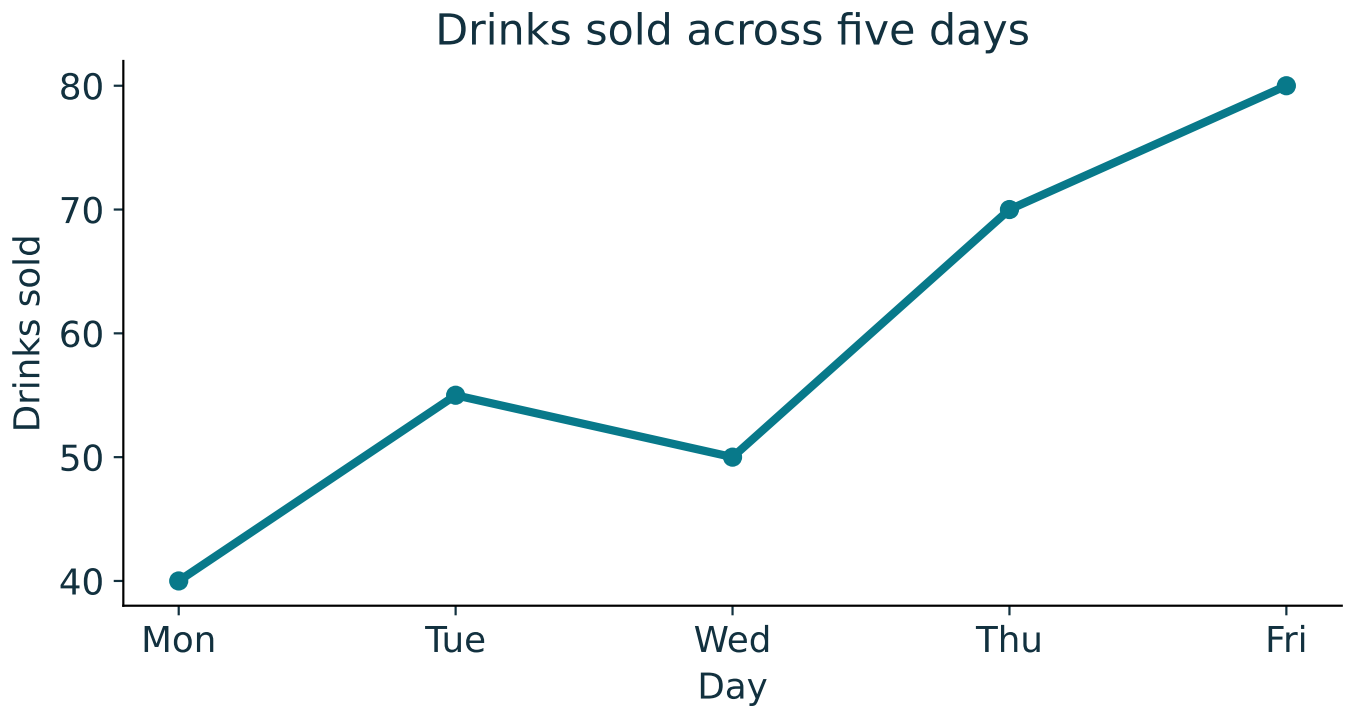
Can count table rows by group and draw bars.

We can use Matplotlib alone. Seaborn can count students in each class from a table. Matplotlib can then add the title.

Your data → **Seaborn (optional)** → **Matplotlib** → **Chart**

Line chart: change over time

A cafe writes down how many drinks it sells each day.



Use it for

Values in time order, such as daily sales or the temperature each hour.

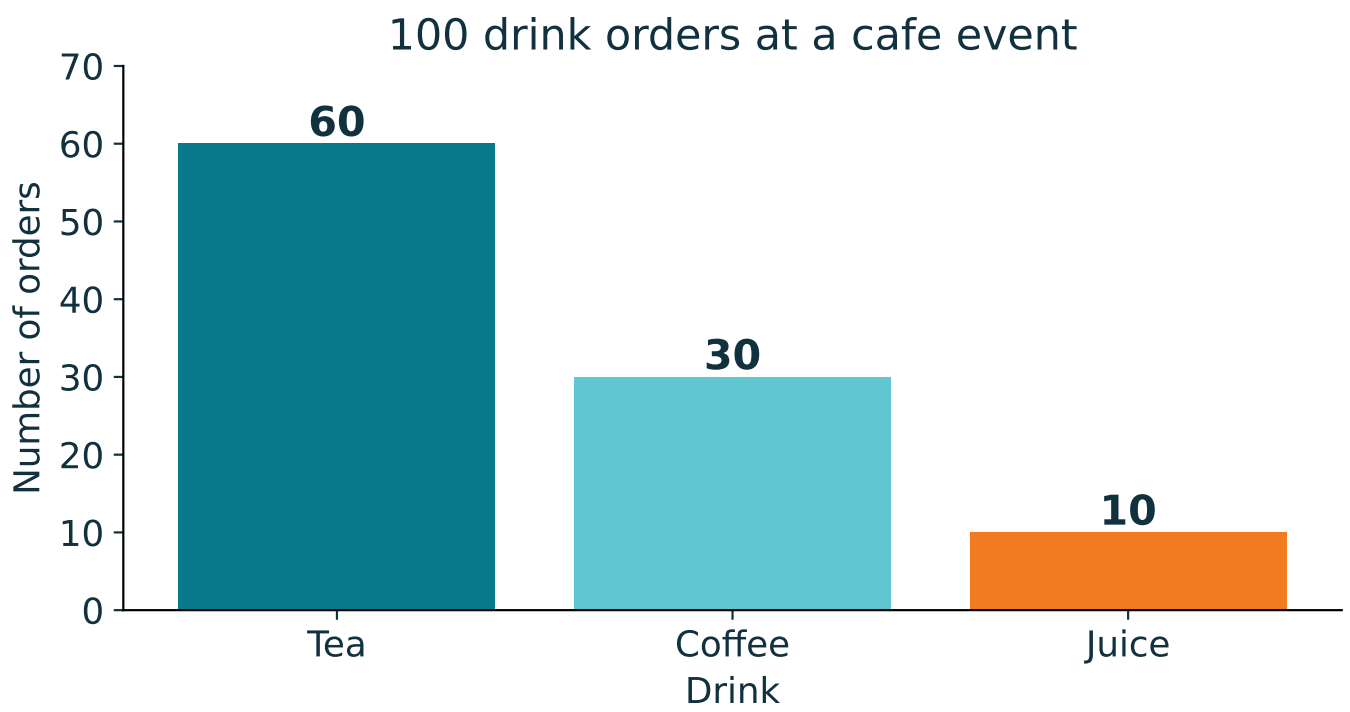
Read this chart

Sales fall from 55 on Tuesday to 50 on Wednesday. Friday has the most sales: 80.

The x-axis runs left to right. The y-axis runs bottom to top. Their labels tell us what we are measuring.

Bar chart: compare named groups

A cafe counts 100 drink orders at an event.



Use it for

Comparing categories. A category is a named group, such as Tea or Coffee.

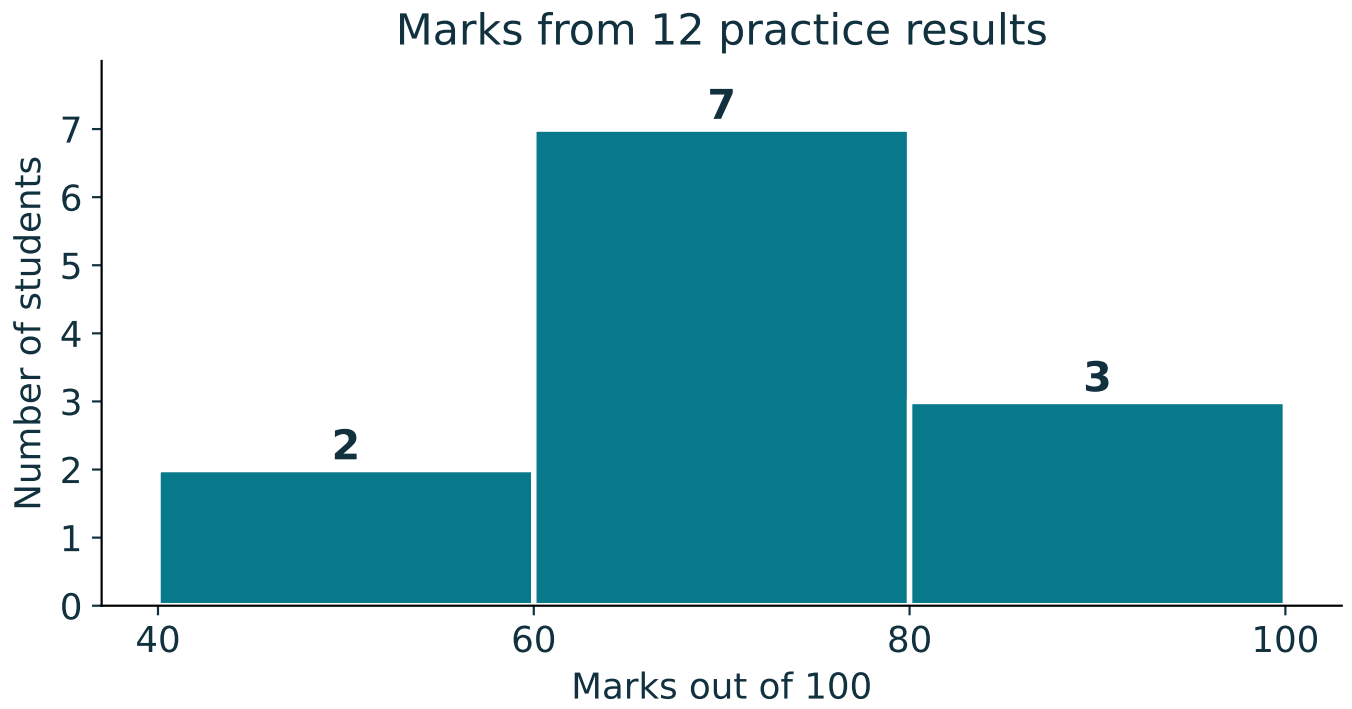
Read this chart

Tea has 60 orders. Coffee has 30. Tea has 30 more orders than Coffee.

Bar or line? Use bars for drink names. Joining Tea to Coffee with a line would suggest a sequence.

Histogram: count marks in ranges

These made-up marks show results for 12 students.



A bin is a number range

40 up to 60 has 2 marks.
60 up to 80 has 7 marks.
80 to 100 has 3 marks.

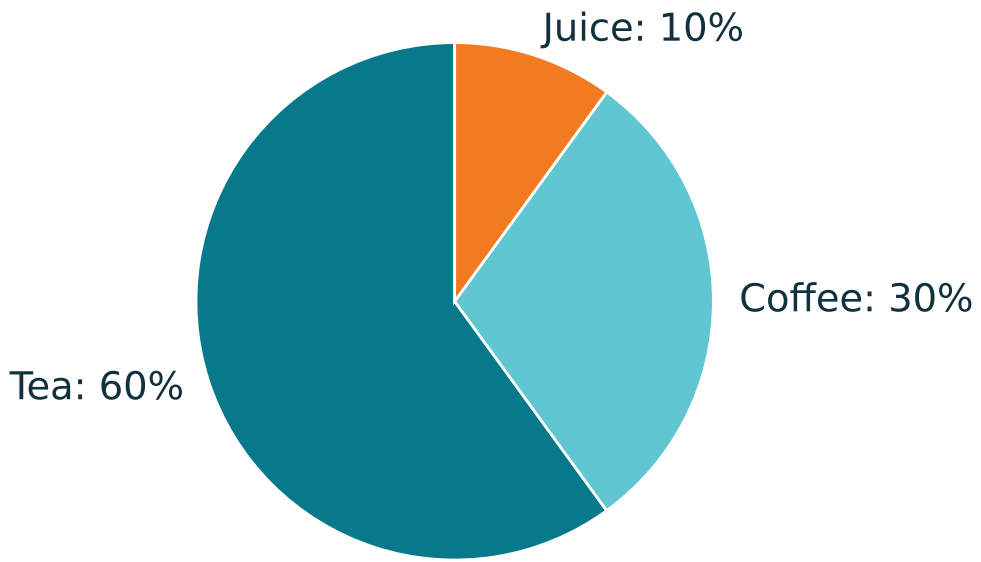
Read this chart

Most marks are from 60 up to 80. The bar height counts students, not their marks.

Bar or histogram? A bar chart compares named groups. A histogram groups numbers into ranges and counts them.

Pie chart: show parts of one whole

We use the same 100 drink orders from the bar chart.



Use it for

A few parts that together make one whole. Each slice is one part of the total.

Read this chart

60 of the 100 orders are Tea, so Tea takes 60% of the pie.

> Pie or bar? Use a pie for the share of the total. Use bars when you want to compare amounts more closely.

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Choose the chart from the question

Start with what you want to find out.

Chart	The question	Everyday example
Line	How did it change over time?	Drinks sold each day
Bar	Which named group has more?	Tea, Coffee and Juice orders
Histogram	Which number range is common?	Marks grouped into ranges
Pie	How is one whole split?	Each drink's share of all orders

Same data, different question: bars compare 60, 30 and 10 orders. A pie shows their 60%, 30% and 10% shares.

DAY 04 · COMPLETE

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